

SIT307 — Machine Learning

Task 11.1HD — Reproducing and Improving a Published Machine Learning Study

A Reproducibility Audit of "An Efficient Stacking-Based Ensemble Technique for Early Heart Attack Prediction", and a Leakage-Aware, Externally Validated Alternative

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Links

Code repository (GitHub): <https://github.com/Ayaan435/heart-disease-reproducibility-sit307>

Video presentation (YouTube, unlisted): <https://youtu.be/Gs2coDPZy4Q>

Paper reproduced: M. Bhagat, A. Sharma and P. Agarwal, *Multimedia Tools and Applications*, vol. 84, pp. 36351–36375, 2025 [1]

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1. Introduction

Machine learning is increasingly proposed as a decision-support tool for the early detection of heart disease, and published accuracies in this area are often extremely high. Bhagat et al. [1] report a stacking ensemble that reaches 98.53% accuracy on a public heart disease dataset, outperforming six individual classifiers and a range of earlier published methods. Results at this level matter, because a model that appears to be almost perfect invites clinical trust and discourages further scrutiny.

This report reproduces that study and examines whether its reported performance survives an independent evaluation. Part 1 re-implements all seven models described in the paper and compares them against the published results, metric by metric. Part 2 proposes and evaluates an alternative methodology motivated directly by what the reproduction uncovers. The central finding is that the dataset used in [1] contains a very large proportion of duplicated patient records, so that almost every record in the test set has already been seen during training. Once those duplicates are removed, the reported advantage of the ensemble disappears entirely. The proposed method therefore focuses not on a new classifier, but on an evaluation protocol that cannot be inflated in this way, combined with

external validation across three independent hospital databases and a decision threshold chosen for the clinical cost of a missed diagnosis.

2. Paper Review

2.1 Problem and motivation

Cardiovascular disease is the leading cause of death worldwide, and early identification of at-risk patients allows preventative treatment before an acute event occurs [1]. Conventional risk scores such as Framingham were developed on population-level data and are known to be imprecise for individual patients, which motivates data-driven alternatives. Bhagat et al. [1] address this by predicting the presence of heart disease from thirteen routinely recorded clinical attributes, including age, sex, chest pain type, resting blood pressure, cholesterol, maximum heart rate achieved and exercise-induced angina.

2.2 Method

The paper follows a conventional supervised learning pipeline. Missing values are imputed using a regression-based method, numerical attributes are normalised and categorical attributes are encoded. The data is then split into training and testing sets, and six classifiers are trained: logistic regression, decision tree, random forest, extreme gradient boosting, naive Bayes and k-nearest neighbours. In the second stage, the predictions of these six models become the input features for a meta-classifier in a five-fold stacking ensemble, which the authors present as their proposed method. Performance is assessed using accuracy, precision, recall, specificity, F1-score, Matthews correlation coefficient and area under the ROC curve.

2.3 Reported results

The authors report that random forest and decision tree are the strongest individual models at 92.68% accuracy, followed by extreme gradient boosting at 90.73%. The stacking ensemble reaches 98.53% accuracy with a precision of 1.0000 and an AUC of 0.988, which the paper presents as outperforming all of the prior work surveyed in its Table 1.

2.4 Critical observations

Three concerns are apparent before any experiment is run.

The dataset is redistributed rather than original. The paper sources its data from a Kaggle redistribution described as containing 1025 patients [1]. The underlying UCI Cleveland database contains 303 patients [2]. The paper does not account for this discrepancy, nor does it report any check for duplicate records. Since the models are evaluated on a random hold-out split, any duplication would place copies of training records into the test set, a well-documented form of data leakage that inflates measured performance without improving the model [3].

The reported metrics are internally inconsistent. Table 12 of [1] reports a Matthews correlation coefficient of exactly 1.0000, which requires a perfect confusion matrix, yet Fig. 5 of the same paper shows three misclassified patients. In Table 11, the Sensitivity column reproduces the Accuracy column for every model, Specificity reproduces F1-score and MCC reproduces Precision, although sensitivity and recall are by definition the same quantity and the two columns differ.

Evaluation is single-split and threshold-agnostic. Performance is reported from one hold-out split with no repetition, so no estimate of variance is available, and with a default decision threshold of 0.5.

In a clinical screening context, a false negative is considerably more costly than a false positive, so accuracy at a fixed threshold is a poor proxy for clinical usefulness [4].

2.5 Research gap

The authors identify two directions for future work: applying the method to a larger dataset, and investigating feature selection techniques [1]. To these this report adds a third and more fundamental gap. The paper describes its source as combining the Cleveland, Hungarian, Switzerland and Long Beach VA databases, yet all training and evaluation is performed within a single population. No evidence is presented that the model transfers to a hospital other than the one that produced the training data, which is the property that matters if such a system is ever deployed [5]. The proposed method in Section 6 addresses all three.

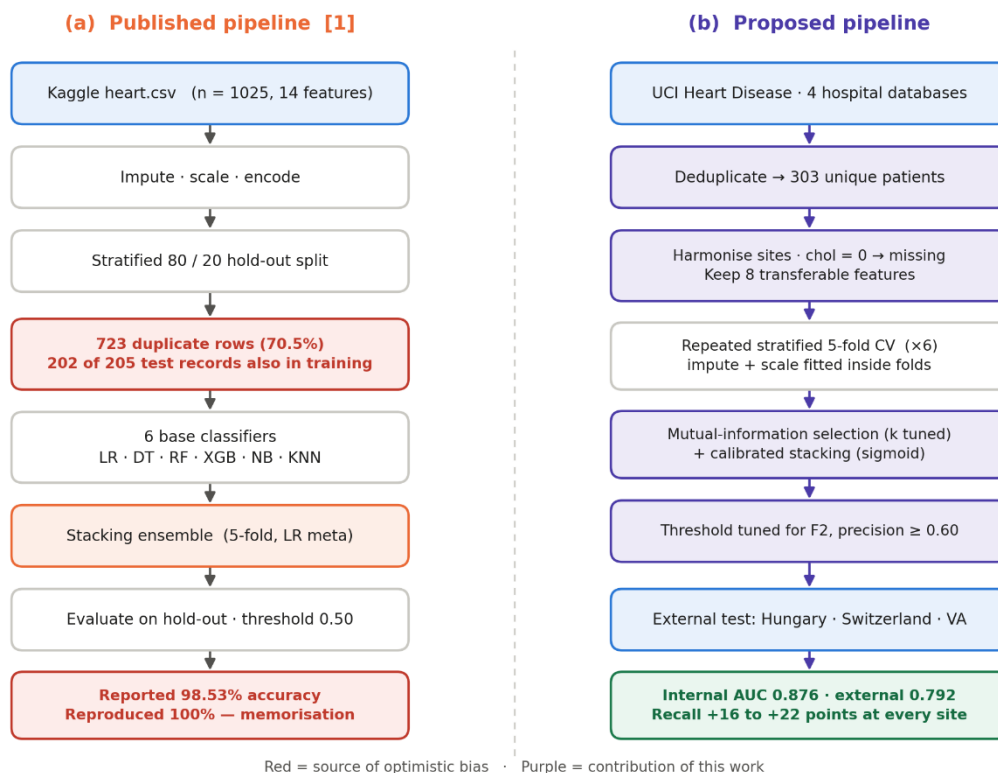


Fig. 1. Comparison of the published pipeline [1] and the proposed pipeline. Red marks the sources of optimistic bias; purple marks the contributions of this work.

3. Dataset and Features

The study [1] uses a Kaggle redistribution of the UCI heart disease data [6], described as containing 1025 patients and 14 attributes. The attributes are the standard Cleveland set: age, sex, chest pain type (cp), resting blood pressure (trestbps), serum cholesterol (chol), fasting blood sugar (fbs), resting electrocardiographic results (restecg), maximum heart rate achieved (thalach), exercise-induced angina (exang), ST depression (oldpeak), the slope of the peak exercise ST segment (slope), the number of major vessels coloured by fluoroscopy (ca), thalassemia (thal), and a binary target.

I obtained the same file from the link given in Section 4 of [1]. It contains 1025 rows and 14 columns with no missing values and a near-balanced target (526 positive, 499 negative). However, only **302**

rows are unique: the file contains **723 exact duplicate records**, a duplication rate of 70.5%. For comparison, the original UCI Cleveland database [2] contains **303 unique patients and no duplicates**, which indicates the duplication was introduced during redistribution rather than existing in the clinical source.

For Part 2 I additionally use the three remaining databases from the same UCI collection: Hungary (294 patients), Switzerland (123) and Long Beach VA (200). Table 1 summarises the four sites.

Table 1. The four UCI heart disease databases.

Database	Patients	Positive rate	Notable data quality issues
Cleveland	303	45.9%	Essentially complete
Hungary	294	36.1%	ca 99.0% missing, thal 90.5%, slope 64.6%
Switzerland	123	93.5%	All 123 cholesterol values recorded as 0; ca 95.9% missing
Long Beach VA	200	74.5%	ca 99.0% missing, thal 83.0%, trestbps 28.0%

Two observations follow. First, the positive rate varies from 36% to 94% across sites, so any model trained at one site faces substantial prior shift at the others. Second, the three attributes obtained through invasive angiography (ca, thal and slope) are almost entirely absent outside Cleveland, yet these are among the highest-ranked features in the paper's own importance analysis (Fig. 2 of [1]). A model that depends on them is therefore not testable outside the hospital that collected them.

4. Reproduction Methodology

All experiments were implemented in Python 3 using scikit-learn 1.6.1 [7] and XGBoost [8], executed in Google Colab. A fixed random seed of 42 was used throughout, and the complete notebook is available in the accompanying repository.

The paper does not report hyperparameters, a random seed, or the choice of meta-classifier, so the following assumptions were made and are stated explicitly:

- **Meta-classifier:** logistic regression. This is the standard choice for stacked generalisation [9] and is the least prone to overfitting the out-of-fold predictions of the base learners.
- **Hyperparameters:** scikit-learn defaults, with 100 trees for random forest and $k = 5$ for k -nearest neighbours, these being the library defaults the paper's description implies.
- **Preprocessing:** standardisation applied to logistic regression, naive Bayes and k -nearest neighbours, which are scale-sensitive, and omitted for the tree-based models, which are invariant to monotone rescaling.
- **Split:** stratified 80/20 hold-out. This is consistent with the paper's confusion matrices, which sum to 205–207 test records, approximately 20% of 1025.

Seven models were then evaluated exactly as in [1]: the six base classifiers and the five-fold stacking ensemble. Seven metrics were computed: accuracy, precision, recall/sensitivity, specificity, F1-score, Matthews correlation coefficient and AUC.

To test whether any discrepancy was attributable to the duplicate records rather than to implementation differences, the identical pipeline was then re-run on the deduplicated data, and both configurations were repeated across 30 random splits to obtain distributions rather than point estimates.

5. Reproduction Results and Critical Analysis

5.1 Reproducing the published results

Table 2 compares my results against those reported in [1].

Table 2. Reproduction of Table 11 of [1] (accuracy).

Model	Reported [1]	Reproduced	Δ
Logistic Regression	0.8439	0.8098	-0.0341
Naive Bayes	0.8439	0.8293	-0.0146
KNN	0.8585	0.8634	+0.0049
Decision Tree	0.9268	0.9854	+0.0586
Random Forest	0.9268	1.0000	+0.0732
XGBoost	0.9073	1.0000	+0.0927
Proposed (Stacking)	0.9853	1.0000	+0.0147

The reproduction is close for the three low-capacity models, within 3.4 percentage points, which confirms the pipeline was re-implemented faithfully. For the high-capacity models the agreement is poor in an unusual direction: **my results are better than the published ones**, with random forest, XGBoost and the stacking ensemble all achieving perfect scores on every metric. A reproduction that outperforms the original is a warning sign rather than a success, because a perfect classifier on 205 unseen patients is not a plausible clinical result.

5.2 Identifying the cause

The explanation is in the data rather than the models. Of the 205 records in the test split, **202 (98.5%) also appear in the training set as exact duplicates**. A sufficiently flexible model can memorise the training records and then reproduce their labels when the same records reappear at test time, which registers as near-perfect accuracy without any generalisation having occurred.

To confirm this, the identical pipeline was re-run after removing duplicates (Table 3, Fig. 2).

Table 3. Effect of removing duplicate records (accuracy, single stratified split).

Model	With duplicates	Deduplicated	Change
Logistic Regression	0.8098	0.8033	−0.0065
Naive Bayes	0.8293	0.7869	−0.0424
KNN	0.8634	0.7869	−0.0765
Decision Tree	0.9854	0.8033	−0.1821
Random Forest	1.0000	0.7541	−0.2459
XGBoost	1.0000	0.7213	−0.2787
Proposed (Stacking)	1.0000	0.7541	−0.2459

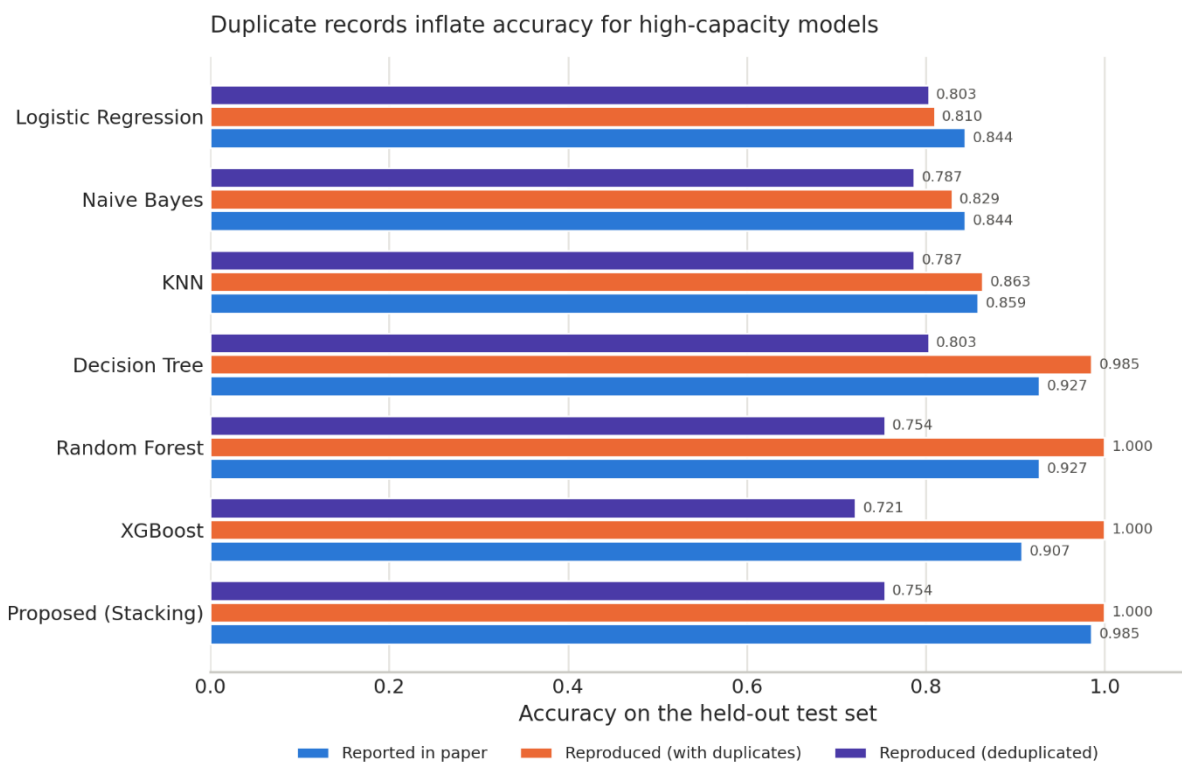


Fig. 2. Effect of removing duplicate records on test accuracy. Logistic regression is essentially unchanged, while the models capable of memorising individual records lose 18 to 28 percentage points.

The pattern is diagnostic. Logistic regression, which fits a single linear decision boundary and cannot memorise individual records, is essentially unchanged (−0.0065). The models capable of memorising lose between 18 and 28 percentage points. Nothing about the data changed other than the removal of repeated rows, so no alternative explanation is available: the reported performance of the ensemble is a measurement artefact.

5.3 Quantifying the effect

Because a single split can be unrepresentative, both configurations were repeated across 30 random stratified splits (Table 4, Fig. 3).

Table 4. Accuracy across 30 stratified splits (mean $\pm$ standard deviation).

Model	With duplicates	Deduplicated
Logistic Regression	0.8372 ± 0.0244	0.8377 ± 0.0425
Naive Bayes	0.8161 ± 0.0200	0.8148 ± 0.0444
KNN	0.8472 ± 0.0209	0.8295 ± 0.0403
Decision Tree	0.9917 ± 0.0099	0.7601 ± 0.0505
Random Forest	0.9953 ± 0.0095	0.8191 ± 0.0478
XGBoost	0.9948 ± 0.0096	0.8131 ± 0.0372
Proposed (Stacking)	0.9948 ± 0.0085	0.8388 ± 0.0373

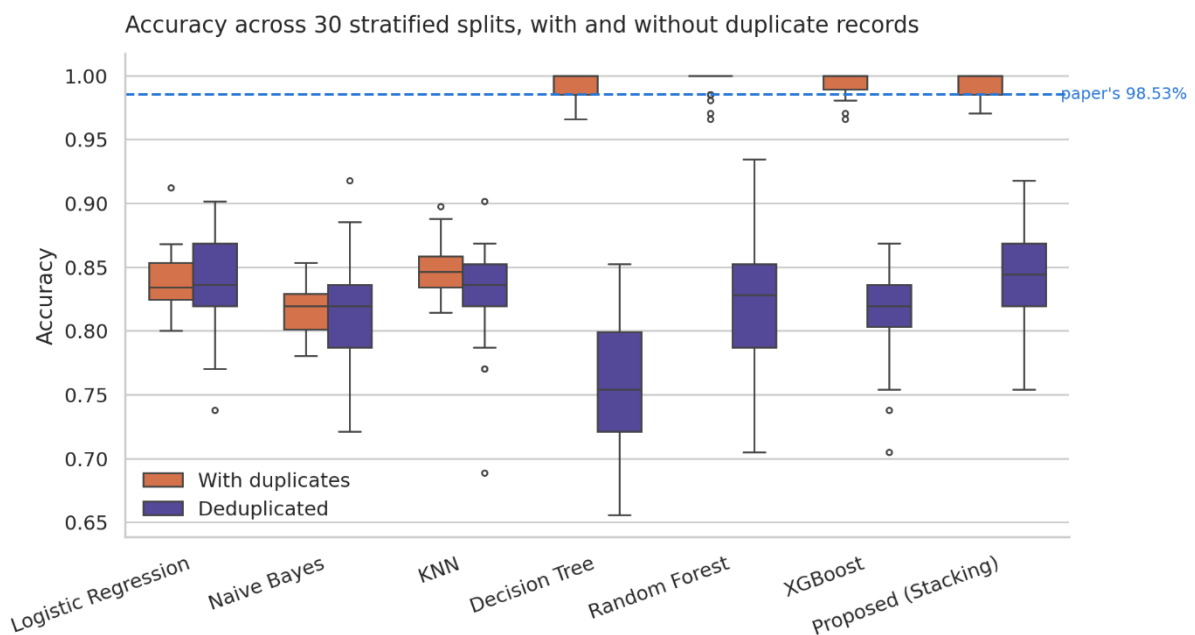


Fig. 3. Distribution of test accuracy across 30 random stratified splits, with and without duplicate records. The dashed line marks the 98.53% reported in [1], which lies 3.9 standard deviations above the deduplicated distribution.

Two findings emerge. First, logistic regression and naive Bayes are unaffected to within one thousandth, confirming that duplication benefits only models with the capacity to memorise. Second, and more consequentially for the paper's contribution, the honest accuracy of the stacking ensemble (0.8388 ± 0.0373) is statistically indistinguishable from that of a single logistic regression (0.8377 ± 0.0425): the mean paired difference is $+0.0011$, with a paired t-test giving $t = 0.311$, $p = 0.758$, and a Wilcoxon signed-rank test giving $p = 0.818$. The reported 98.53% lies **3.9 standard deviations** above this distribution.

The central claim of [1], that stacking "considerably enhances" the performance of the base classifiers, is therefore not supported once the evaluation is corrected. The apparent improvement is entirely attributable to duplicate records interacting with a random hold-out split.

5.4 Reporting inconsistencies

Two verifiable inconsistencies appear in the reported results. Table 12 states an MCC of 1.0000 for the proposed model, but the confusion matrix in Fig. 5 of [1] records 95 true negatives, 0 false positives, 3 false negatives and 107 true positives. Recomputing from those counts gives an MCC of 0.9711, not 1.0000. The same recomputation shows that Table 12's reported sensitivity (0.9853) is in fact the accuracy, and its reported specificity (0.9861) is in fact the F1-score; the true values are 0.9727 and 1.0000 respectively.

This is not an isolated transcription error. In Table 11 of [1], the same three substitutions hold for all seven models: the sensitivity column is identical to the accuracy column, the specificity column is identical to the F1-score column, and the MCC column is identical to the precision column, in every row. The consistency of the pattern across both tables indicates a systematic fault in the metric-reporting code rather than a typographical slip, and it means that three of the eight reported metrics do not measure what their labels claim.

6. Proposed Method

6.1 Motivation

The reproduction in Section 5 shows that the reported performance of [1] is an artefact of the evaluation protocol rather than a property of the model. This reframes what an improvement should be. Proposing a different classifier would not address the problem, because the flaw is not in how the ensemble learns but in how it was measured. The proposed method is therefore a **protocol**, with four components, each answering a specific weakness identified above and two of which correspond directly to the future work the authors themselves nominate.

6.2 Design

(i) Duplicate-aware validation. Exact duplicate records are removed before any split; in the redistributed file this reduces 1025 rows to 302 unique records. Part 2 is therefore conducted on the original UCI Cleveland database (303 patients), which contains no duplicates, and evaluation uses repeated stratified five-fold cross-validation rather than a single hold-out. This removes the leakage pathway entirely and reports variance rather than a single number.

(ii) Cross-site feature harmonisation. Cholesterol values of 0, which are coded missing values rather than measurements, are converted to missing. Features are then restricted to those recorded in at least half of the patients at every external site, which retains eight: age, sex, cp, trestbps, restecg, thalach, exang and oldpeak. This excludes ca, thal and slope, which rank highly in the paper's importance analysis but are almost entirely absent outside Cleveland. The restriction costs internal accuracy, but it is the only configuration under which external validation is possible at all.

(iii) Feature selection and probability calibration. Mutual-information selection [10] is applied with k tuned over $\{4, 6, 8\}$ by internal cross-validation, addressing the feature-selection direction nominated in [1]. The stacking ensemble is then wrapped in sigmoid calibration [11], so that its outputs can be interpreted as probabilities rather than arbitrary scores.

(iv) Cost-sensitive decision threshold. The paper evaluates at the default threshold of 0.5, which implicitly assumes a false negative and a false positive are equally costly. In screening for heart disease they are not: a missed case may be fatal, while a false alarm leads to further testing. The threshold is therefore chosen to maximise the F2-score, which weights recall twice as heavily as precision, computed on out-of-fold predictions and constrained to precision ≥ 0.60 so that the degenerate "classify everyone as positive" solution is excluded. The selected threshold was **0.21**, with an out-of-fold F2 of 0.848 at precision 0.629.

(v) External validation. The final model, trained only on Cleveland, is tested on the Hungarian, Swiss and Long Beach VA databases, which it has never seen. This addresses the "larger dataset" direction in [1] in the most informative way available: not more data from the same hospital, but data from different ones.

7. Experimental Protocol

Two configurations were compared on identical data and features, so that the effect of the proposed components is isolated:

- **Configuration C (baseline):** the stacking ensemble of [1], trained on deduplicated Cleveland using the eight transferable features, evaluated at the default 0.5 threshold.
- **Configuration D (proposed):** the same ensemble with mutual-information selection, sigmoid calibration and the tuned decision threshold.

Internal performance was estimated by repeated stratified five-fold cross-validation with six repeats (30 folds). External performance was measured by fitting on the full Cleveland set and predicting each external database once. Metrics reported are accuracy, precision, recall, F1, F2, AUC, Matthews correlation coefficient and the Brier score. AUC is emphasised in the comparison because it is independent of the decision threshold and therefore separates the quality of the model's ranking from the choice of operating point.

8. Results and Comparative Analysis

8.1 Internal performance

On deduplicated Cleveland with eight features, the proposed pipeline achieves the results in Table 5.

Table 5. Internal performance, 30 folds (mean $\pm$ standard deviation).

Metric	Value
Accuracy	0.807 $\pm$ 0.047
Precision	0.828 $\pm$ 0.077
Recall	0.739 $\pm$ 0.066
F1	0.778 $\pm$ 0.054
F2	0.753 $\pm$ 0.059
AUC	0.876 $\pm$ 0.037

MCC	0.614 ± 0.097
Brier score	0.145 ± 0.023

Label-based metrics are computed at the default 0.5 threshold; AUC and the Brier score are threshold-free. The tuned threshold of 0.21 described in section 6.2(iv) is applied in the external evaluation of Table 6, where the cost asymmetry it targets is the object of study.

This is 18 percentage points below the 98.53% reported in [1], and is the honest figure for this dataset. It is also achieved with five fewer features, three of which are obtained invasively.

8.2 External validation

Table 6 reports performance at the three hospitals never seen during training.

Table 6. External validation: trained on Cleveland, tested elsewhere.

Test site	Configuration	Accuracy	Recall	Precision	F1	F2	AUC
Hungary	C: baseline	0.830	0.736	0.780	0.757	0.744	0.879
	D: proposed	0.731	0.896	0.583	0.706	0.809	0.884
Switzerland	C: baseline	0.724	0.739	0.955	0.833	0.774	0.775
	D: proposed	0.919	0.957	0.957	0.957	0.957	0.760
Long Beach VA	C: baseline	0.720	0.805	0.816	0.811	0.808	0.727
	D: proposed	0.760	0.973	0.767	0.858	0.924	0.732

The proposed configuration improves recall at every site, by 16.0, 21.8 and 16.8 percentage points respectively, and improves F2 at every site. At Switzerland it improves every metric simultaneously. At Hungary the gain in recall is paid for in precision, which falls from 0.780 to 0.583: in a screening context this trade is defensible, since the cost is additional follow-up testing rather than a missed diagnosis, but it should be stated rather than hidden behind an aggregate score.

Note that AUC is almost unchanged between the two configurations ($0.879 \rightarrow 0.884$, $0.775 \rightarrow 0.760$, $0.727 \rightarrow 0.732$). This is an honest and important qualification: **calibration and threshold tuning change the operating point, not the underlying ranking of patients**. The contribution is a decision rule appropriate to the clinical cost asymmetry, not a model that discriminates better.

8.3 Generalisation and calibration

Mean external AUC is 0.792 against an internal 0.876, a generalisation gap of 0.084. Performance degrades with distance from the training population: Hungary, whose case mix is closest to Cleveland, retains an AUC of 0.884, while the VA database falls to 0.732.

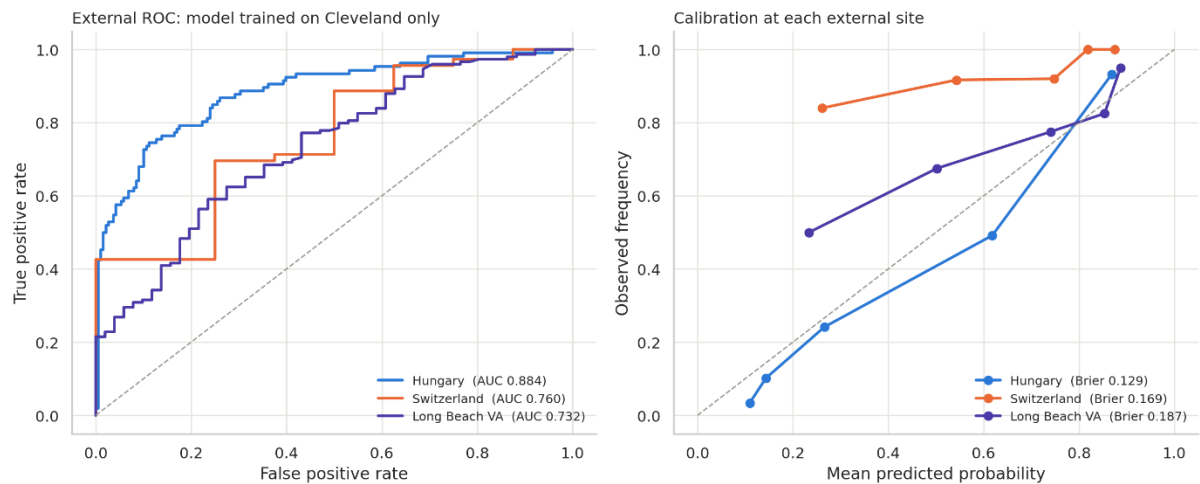


Fig. 4. External validation of the proposed pipeline. Left: ROC curves at the three hospitals never seen during training. Right: calibration curves, showing systematic under-prediction at Switzerland and over-prediction at Hungary.

Fig. 4 shows why threshold and calibration must be revisited per site. The Swiss cohort is 93.5% positive against Cleveland's 45.9%, and the calibration curve for Switzerland sits well above the diagonal, meaning the model systematically under-predicts risk there. Hungary, at 36.1% positive, sits below the diagonal and over-predicts. Brier scores are 0.129, 0.169 and 0.187 for Hungary, Switzerland and VA respectively. A model transferred between hospitals therefore needs its calibration re-fitted on local data even when its ranking remains useful, a form of prior shift that is invisible to accuracy alone [5].

8.4 Comparison with the published method

Table 7. Summary of the two approaches.

	Published [1]	Proposed
Duplicate records	Not checked; 70.5% present	Removed before splitting
Evaluation	Single 80/20 hold-out	Repeated stratified 5-fold ($\times 6$)
Features	13, including 3 site-specific	8 transferable, selected by mutual information
Probabilities	Uncalibrated	Sigmoid-calibrated, Brier reported
Threshold	0.50 default	0.21, tuned for F2 with precision ≥ 0.60
External validation	None	3 independent hospitals
Headline accuracy	98.53%	80.7% internal, 73.1–91.9% external

The proposed method reports a substantially lower headline number, and that is the point. A figure of 0.807 that survives deduplication, repeated cross-validation and out-of-hospital testing is more informative than 0.9853 obtained on records the model has already seen.

9. Conclusion and Future Work

This report reproduced the stacking ensemble of [1] and found that its reported 98.53% accuracy is not attributable to the method. The dataset contains 723 duplicate records, and 98.5% of the test set also appears in training, so high-capacity models score highly by recognising records rather than generalising from them. Removing the duplicates reduces the ensemble to 0.8388 ± 0.0373 across 30 splits, statistically indistinguishable from a single logistic regression ($p = 0.758$), while logistic regression itself is unaffected by the duplicates, confirming the mechanism. Two internal inconsistencies in the published results were also identified, both traceable to the same fault in the metric-reporting code.

The proposed protocol substitutes leakage-aware evaluation, cross-site feature harmonisation, calibrated probabilities and a cost-sensitive threshold, and adds the external validation the original study lacks. It reports 0.807 internal accuracy and 0.792 mean external AUC, and raises recall by 16 to 22 points at every external hospital, at the cost of precision at one of them.

Three limitations should be acknowledged. The training set is small at 303 patients, so confidence intervals are wide. External evaluation was performed once per site rather than repeated, since each site is a fixed cohort. And the eight-feature restriction, while necessary for transferability, discards information that is genuinely predictive where it is available.

Future work follows directly. Site-specific recalibration should be evaluated, given the prior shift evident in Fig. 4. Multi-site training, for instance leave-one-hospital-out cross-validation across all four databases, would test whether pooling improves transfer. Finally, the duplication found here is a property of a widely used public file rather than of one paper, so a systematic audit of published results that rely on the same redistribution would be a worthwhile contribution in itself.

10. GenAI Acknowledgement

I used several AI tools during this task, and all uses are declared here. Claude (Anthropic) [12] was used as a coding and writing assistant: to help structure and debug the reproduction notebook, to suggest the statistical tests and evaluation protocol used in Sections 5 and 7, to generate the matplotlib code for Figs. 1–4, and to help draft and edit this report. Gemini [13], accessed through Google Colab, was used for minor code corrections. Microsoft Copilot [14] was used for formatting assistance and word-count estimation, and QuillBot [15] was used to refine wording in some sections.

All experiments were designed, executed and verified by me. Every number reported in Tables 2–7 was produced by the notebook in the accompanying repository and can be reproduced by re-running it. The identification of the duplicate records in the dataset, the inconsistencies in the published results tables, and the interpretation of all findings are my own. All AI-assisted text was reviewed and edited before inclusion. Two errors introduced during drafting were caught in this review: an incorrect patient count in Section 6.2, which conflated the 302 unique records in the redistributed file with the 303 in the original UCI database, and a reference list in which access dates had been applied to print sources that do not require them.

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